

Replace a Method Invocation

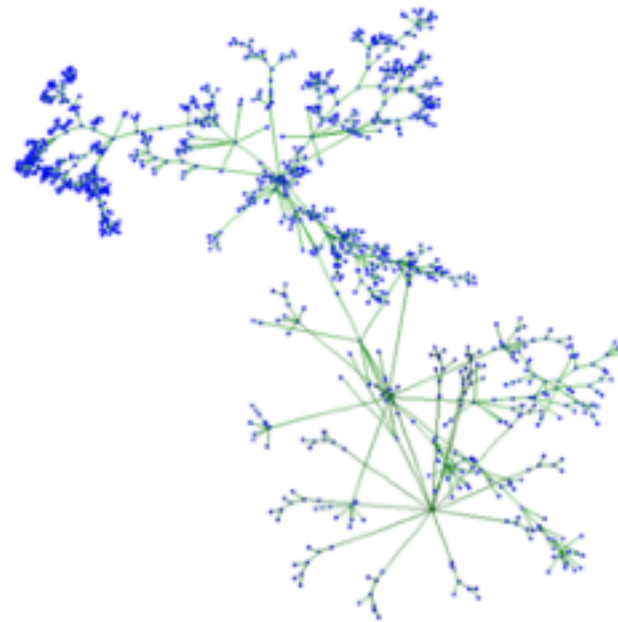
```
@RecipeDescriptor(  
    name = "Replace oldOperation with betterOperation",  
    description = "Replace deprecated `oldOperation` " +  
                  "with surrogate `betterOperation`")  
public static class ReplaceOldOperation {  
    @BeforeTemplate  
    public void oldOperation(SomeService s) {  
        s.oldOperation();  
    }  
    @AfterTemplate  
    public void newOperation(SomeService s) {  
        s.betterOperation();  
    }  
}
```

100% accurate, style-preserving transformation because of a tech leap in code representation

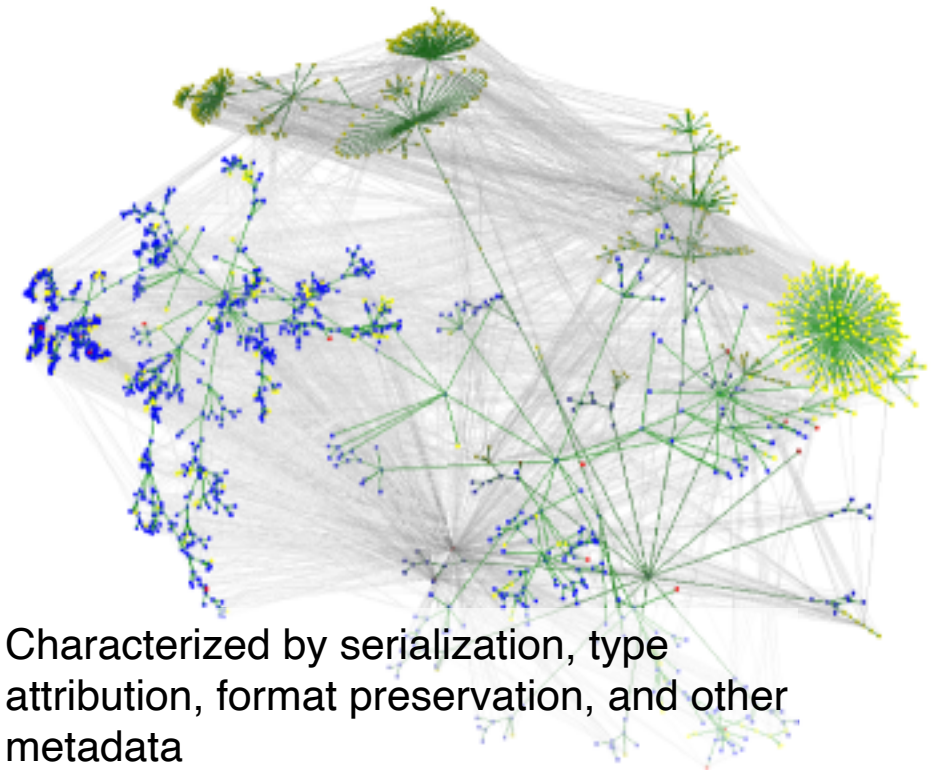
Text-based search

```
void m() {  
    if(c > 2) {  
        x = c;  
    }  
    while(c < 10) {  
        x += p();  
        c++;  
    }  
}
```

Abstract Syntax Tree (AST)



Lossless Semantic Tree (LST)



Characterized by serialization, type attribution, format preservation, and other metadata

+ another 600,000 nodes that didn't fit in this graphic